

Review Article

Advanced precision veterinary technologies and smart boluses: Innovations in drug delivery, health monitoring, and future perspectives

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ABSTRACT

Advancements in veterinary medicine are revolutionizing animal health management through precise drug delivery and real-time health monitoring, significantly improving livestock health and productivity. Smart bolus is designed to provide controlled, sustained release of therapeutic agents, making them highly effective for managing the health of commercial animals. Recent innovations, including advanced sensors, wireless communication, biodegradable materials, and 3D-printed boluses, enable real-time monitoring of critical parameters such as temperature, pH, and drug concentrations within the animal. These technologies find applications in early disease detection, heat stress management, feed optimization, and reproductive health, greatly enhancing animal welfare and farm profitability. Furthermore, the integration of IoT, artificial intelligence based analytics, and blockchain-based traceability has improved the functionality and reliability of smart boluses, driving advancements in precision livestock farming. By ensuring accurate dosing, reducing handling frequency, and maintaining consistent therapeutic levels, these devices enhance treatment outcomes. Additionally, they autonomously monitor metabolic health in ruminants/non-ruminants and deliver targeted treatments for gastrointestinal infections, such as antibiotics or nutrients, minimizing the need for repeated manual interventions. This review explores the current advancements in smart bolus technology, highlights its transformative applications, and evaluates its potential to foster sustainable and efficient livestock management systems.

1. Introduction

As the global population expands, the reliance on livestock to satisfy essential nutritional demands has introduced a paradox: increasing consumption of animal-based foods exerts considerable strain on environmental resources [1,2]. In recent years, awareness of animal welfare has intensified as society addresses the ethical considerations associated with animal treatment [3]. Despite these challenges, veterinary care remains an area with significant potential for positive impact. One promising avenue is the development and distribution of veterinary boluses and advanced devices, a technology that enhances animal health management and meets the evolving needs of the industry [4,5]. The use of bolus has long been central to effective health management in

veterinary medicine, especially for large livestock such as cattle, horses, and sheep [6]. A bolus, typically a large tablet for oral administration, offers a practical and efficient way to deliver precise doses of medications, nutrients, and therapeutic agents over an extended period [4,7]. Administered using specialized tools like balling guns, boluses ensure active ingredients reach the stomach or rumen, allowing for gradual dissolution and absorption. This controlled release capability reduces the frequency of handling and administration, lowering stress for both animals and caretakers [8].

Advancements in bolus/devices technology represent a significant step forward in the management and treatment of livestock. Traditional boluses, known for their ability to deliver accurate doses of medications and nutrients, are now being upgraded with advanced technologies that

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provide deeper insights and greater control over animal health [9]. Numerous studies highlight how smart boluses/devices, equipped with sensors and connectivity features, facilitate continuous monitoring of physiological parameters in livestock [10]. This data is often transmitted in real-time to a central platform, allowing veterinarians, farmers, and caretakers to track individual animals within large herds and detect early signs of illness or nutritional imbalances [11]. In addition, the smart boluses enable the controlled release of drugs, offering precise, timed delivery that enhances treatment effectiveness, reduces the frequency of administration, and ensures long-term health management [12]. These devices can also detect changes in body physiology, which may indicate the onset of infections or digestive issues, allowing for timely intervention and potentially reducing the need for extensive treatments [13]. By incorporating such innovative technology, veterinary practices not only improve treatment outcomes but also foster more sustainable and efficient livestock management practices.

In animal care, 3D printing technology is emerging as a transformative tool, enabling the creation of custom-designed boluses tailored to the unique health needs of individual animals. This technology allows for precise control over bolus size and release profiles, optimizing the delivery of medications, nutrients, and supplements. Through 3D printing, veterinarians can enhance treatment effectiveness, promote animal health, improve dose distribution, and reduce waste in livestock management. Previous studies have explored 3D printing in veterinary radiotherapy, underscoring its potential to improve treatment accuracy and efficiency [14]. The smart technology can serve multiple functions, ranging from nutritional boluses that supply essential minerals to therapeutic boluses for targeted treatments, and smart boluses/devices embedded with sensors for real-time health monitoring, which collectively enhance livestock health management. For example, in dairy farming, these devices can track the core body temperature of cows, providing early warnings for heat stress, infections, or other health concerns that may affect milk yield and quality. Smart bolus devices are also useful in monitoring reproductive health, optimizing breeding strategies, and improving herd fertility rates [15].

Advancement in technology offers wide-ranging, impactful applications, particularly in dairy and beef farming, where herd health is crucial for productivity and profitability. These technologies provide substantial benefits by facilitating proactive, precise health management. Additionally, artificial intelligence (AI) holds promise for transforming various aspects of veterinary care, potentially enhancing the accuracy and safety of diagnostics, treatment decisions, and other critical stages when thoughtfully integrated into veterinary practices [16,17]. Conversely, this technology is used to deliver drugs and nutrients, promoting better health and well-being in livestock. A recent study assessed the impact of a supplement range on improving cow body condition score (BCS) and body weight (BW), as well as calf birth, weaning, and post-weaning weights, including weight per day of age (WDA). Moreover, by automating the monitoring process, smart devices alleviate the need for frequent manual checkups, saving time and labor while reducing stress on the animals. This comprehensive review examines recent progress in veterinary technologies, highlighting the development and application of smart boluses, sensor-based wearable devices, and 3D printing, alongside their current roles in livestock management and their significant potential for future innovations. As agriculture embraces sustainable and data-driven approaches, smart technologies represent the forefront of high-tech veterinary care. These innovations enable proactive decision-making, enhance animal welfare, and optimize farm operations. The future of veterinary science is increasingly tied to the integration of these advanced tools, promoting more sustainable, efficient, and precise livestock farming practices that benefit both animals and producers.

2. Conventional method for veterinary treatment

In conventional veterinary practice, a bolus refers to a large solid

tablet with a cylindrical or oval shape, formulated for oral administration in animals such as cattle, horses, and sheep. These boluses are designed to deliver drugs, nutrients, or therapeutic agents over a pre-determined duration, ensuring effective management of various health conditions. [6,18]. The primary function in veterinary practice is to provide a controlled release of active pharmaceutical ingredients (APIs), ensuring a sustained and gradual dispersion of these ingredients. This method enhances treatment efficacy and reduces the need for frequent administration. A recent study demonstrated that administering pharmaceutical compounds through traditional oral boluses typically requires drug quantities in the gram range rather than the milligram range. This study examines the extent to which significant changes in the *in vitro* performance of sulfamethazine boluses can be tolerated before impacting *in vivo* blood level profiles [19]. This placement aids in swallowing and ensures the bolus reaches the rumen directly, where it can dissolve and release active ingredients for systemic absorption. The primary advantage of conventional boluses lies in their ability to treat bacterial infections and parasitic worm infestations through the oral administration of antibiotics and anthelmintic agents, respectively. These tablets are specifically designed to target particular parasites, helping to maintain the health and productivity of livestock [20]. Similarly, the sustained release mechanism of boluses allows for site-specific delivery of antibiotics, thereby reducing the risk of antibiotic resistance by maintaining consistent drug concentrations within the animal [21]. In veterinary medicine, parasitic diseases are a significant health concern. Effective parasitic control and treatment necessitate novel pharmaceutical strategies, incorporating a diverse range of anthelmintic drugs tailored to their biological responses and metabolic pathways. However, the effectiveness of this method depends on the preparation of bolus formulation and administration, as factors such as size, composition, and dissolution rate must be carefully managed to avoid complications like esophageal irritation or incomplete drug release. Overall, traditional methods are a valuable tool in veterinary practice for ensuring the effective and efficient treatment of bacterial infections in livestock, contributing to better animal health and farm productivity [22]. These formulations must ensure the specificity, stability, and efficacy of the API throughout the release period. However, the effective application of boluses requires careful consideration of various factors to prevent discomfort or injury during administration [23]. These Boluses are administered with specialized tools such as balling guns, ensuring that the active ingredients are delivered directly into the stomach or rumen for gradual dissolution and absorption. The conventional methods of administration in veterinary practice present several challenges, including the formation of an air gap between the bolus material and the skin during radiation therapy, which can lead to underdosing of the tumor treatment [24]. Moreover, conventional boluses may not conform well to the irregular surfaces of animal bodies, leading to inconsistent dose distribution and reduced therapeutic efficacy. The necessity for multiple treatments with traditional bolus methods, combined with the significant labor involved, can be both costly and time-consuming. These concerns highlight the importance of continued research and development efforts to enhance the efficacy, reliability, and cost-effectiveness of traditional veterinary practices.

3. Recent technological advancements in veterinary care

Technological advancements in veterinary care systems have greatly improved the monitoring and management of livestock health, leading to increased productivity and operational efficiency (Table 1). The integration of computer systems and software has played a crucial role in this progress, making livestock management more convenient for farmers while increasing efficiency and profitability in animal production. As a result, technological innovations have become a primary focus for enhancing animal productivity and sustainability [37]. Modern equipment and tools have simplified animal husbandry, providing farmers with numerous advantages. Key daily tasks such as milking,

Table 1
Technological advancements in veterinary medicine and practice.

Technology	Description	Applications	References
Diagnostic Advancements			
Precision Imaging	Advanced imaging techniques like digital radiography, ultrasound and CT scan.	- Digital radiography detects fractures and cancers, - Ultrasound examines soft tissues.	[22]
Molecular Diagnostics	Diagnostic tool uses PCR and NGS to detect DNA/RNA-based genetic defects and infectious diseases.	- CT scans the brain, spine, and bones for diagnosis. - PCR diagnoses diseases like parvovirus.	[25]
Sensor Bolus/Device	Sensor boluses are maintenance-free, easy to ingest, and remain in the rumen (reticulum) for continuous use.	- NGS detects genetic mutations in canine influenza. - Device monitors vital parameters like temperature.	[26]
Wearable Technology			
Health Monitoring Device	Devices like smart collars, ear tags, and trackers track vital metrics such as heart rate and movements.	They offer real-time data, enabling early detection of health issues.	[27]
IoT Applications	Large farm owners can use wireless IoT to track cattle location, health, and well-being.	This aid offers precision, real-time feedback.	[28]
Robotics and Automation			
Robotic Rehabilitation	This aid offers precision, real-time feedback, allowing veterinarians to adjust therapy for improved rehabilitation.	Wireless IoT to track cattle location, health, and well-being.	[29]
Surgical Robots	They assist surgeons, enhancing safety, precision, and reducing scars.	Robotic surgery improves outcomes, reduces risk, and lowers stress for animals and vets.	[30]
3D Printing			
Prosthetics and Orthopedics	These devices alleviate pain and improve movement and comfort, enhancing the animal's quality of life.	- Limb prosthetics aid animals with leg injuries or defects, - Orthotics stabilize joints.	[31]
Radiotherapy and Dose Distribution	A 3D-printed bolus offers improved dosimetry over a commercial bolus	3D-printed boluses minimize air gaps. - Improved dosimetry	[32]
Drug Delivery			
Controlled Delivery System	Controlled drug delivery systems transport anticancer agents like doxorubicin, paclitaxel, etc.	- Controlled systems reduce drug toxicity. - Target specific organs or systems.	[33]
Targeted Drug Delivery System	They deliver medications directly to affected areas, providing stronger treatments with fewer side effects.	Targeted chemotherapy increases effectiveness and reduces discomfort.	[34]
Advanced Monitoring Equipment			
Anesthesia Monitoring	These systems continuously monitor animals during surgery, minimizing risks and ensuring safer, more seamless outcomes.	These tools track vital signs in real-time, enabling vets to make immediate adjustments.	[35]
Vital Sign Monitoring	These devices track vital patterns and provide early alerts, helping vets address issues before they worsen.	In ICU and monitoring continuous vital sign tracking ensures timely intervention.	[36]

feeding, environmental control, and monitoring reproductive performance are now more manageable through smart systems, which reduce human error and improve decision-making, ultimately supporting product quality and profitability [38]. The recent study investigates the rise of automation in animal husbandry demands ongoing, sophisticated monitoring of livestock. Smart sensors, a rapidly evolving technology, offer valuable insights into animal health, physiological metrics, and estrus detection. Herein, the development of a new sensor, covering design concepts, sensor types, mechanical and power solutions, communication methods, and field-tested detection algorithms, concludes with current trends and prospects [39]. Moreover, smart drug delivery systems equipped with controlled-release mechanisms now provide a consistent supply of medications and nutrients, optimizing treatment protocols and reducing the need for frequent animal handling in veterinary care. These innovations are revolutionizing veterinary practice by offering more accurate and effective tools for health management, resulting in better outcomes and greater sustainability in livestock farming.

3.1. 3D printer in veterinary medicine

Veterinary drug treatment can be challenging due to differences in size, body mass, and pharmacokinetic variability between species. 3D printing is transforming veterinary medicine, offering innovative ways to enhance animal diagnosis, treatment, and surgical planning. This technology allows veterinarians to produce highly accurate, customized animal anatomy models, enabling them to examine complex cases, plan surgeries, and practice intricate procedures. Additionally, 3D printing reduces surgery time, provides greater precision, and improves patient outcomes [40,41]. In a recent study, researchers investigated the use of three semi-solid extrusion 3D printers in a pharmacy or animal clinic setting for the extemporaneous manufacturing of prednisolone-containing or dispersible films for veterinary use (Fig. 1)

[42]. 3D printing enables the creation of modified veterinary boluses with precise shape and tailored drug dose distribution, improving site-specific delivery in animal care within clinical settings. This technology enables customization of drug release according to the specific needs of each animal, improving treatment accuracy, reducing medication waste, and enhancing efficiency in livestock health management. One of the primary uses of 3D printing in veterinary medicine is the development of prosthetics and implants. For animals with congenital deformities, injuries, or amputations, 3D-printed prosthetics offer a customized fit, restoring functionality and improving quality of life. Additionally, 3D printing allows for rapid prototyping and testing of prosthetic designs, reducing both time and costs [43].

Another major use of 3D printing is in designing customized surgical guides, tools tailored to help surgeons navigate complex anatomical areas with precision. These guides are especially valuable for intricate procedures, like orthopedic surgeries, as they offer exact templates for surgical tasks, thereby reducing errors and the likelihood of repeat operations. [45,46]. In a similar advancement, fused filament fabrication (FFF) has been explored for creating biomedical implants. This study established a CAX-based workflow to produce customized, biocompatible 316L stainless steel implants using FFF. The workflow was applied to two veterinary cases, demonstrating the effectiveness of FFF-processed AISI 316L stainless steel implants for tailored medical applications [47]. Researchers highlighted that orthoses and socket prostheses are rarely used in veterinary orthopedics, yet their demand is increasing despite limited evidence and high costs. In the recent study, 3D scanning and fused deposition modeling were used to develop orthoses and socket prostheses for nine dogs and one calf [44]. Briefly, they used innovative adaptation scales and found that orthoses showed excellent fit, with the calf demonstrating the best and most immediate adaptation among all cases. Apart from this, the commercial boluses, which create gaps and dose discrepancies on uneven surfaces, 3D-printed based boluses improve dose distribution in photon beam

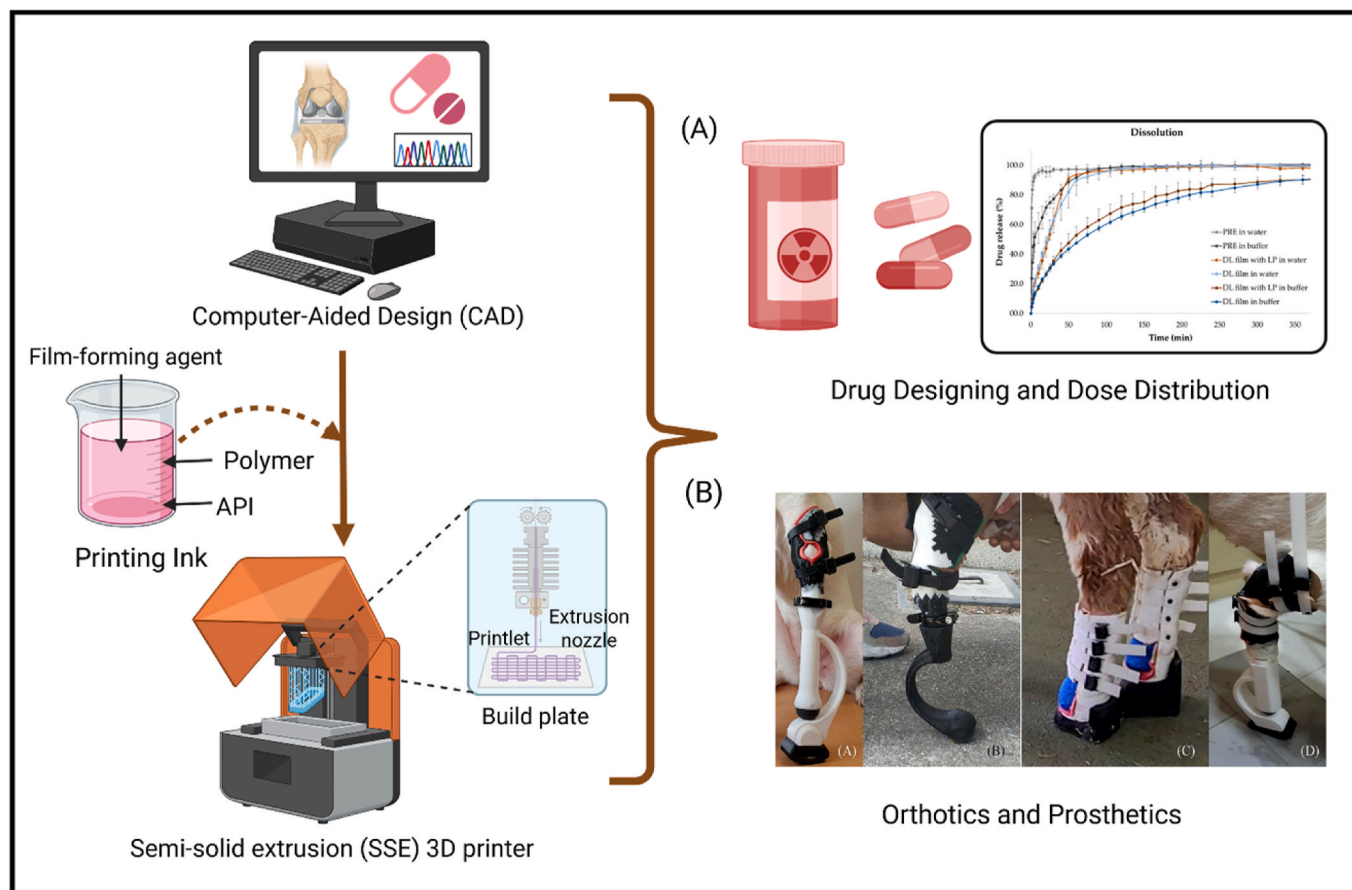


Fig. 1. Schematic of 3D printer-based veterinary medicine applications: (A) Drug design and dose distribution [32]. (B) Applications in prosthetics and orthotics, including (A) Thoracic prosthesis, (B) Thoracic prosthesis, (C) Pelvic prosthesis, and (D) Thoracic prosthesis [44]. [All images (A, B, C) reproduced from the cited paper with kind permission of the copyright holder, an open-access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license, which permits unrestricted use, distribution, and reproduction in any medium] (Created through [BioRender.com](https://www.biorender.com)).

radiation therapy for veterinary patients [32]. Generated within planning software and 3D printed, these devices minimize air gaps, achieve dosimetry closer to the prescription, and are recommended for irregular superficial treatment areas. Overall, 3D printing is advancing the field of veterinary medicine by offering personalized solutions that transform animal healthcare, reduce recovery times, and increase the precision of treatments (Fig. 1).

3.2. Wearables technology

Wearable technology is revolutionizing veterinary care by enabling a more proactive and individualized approach to animal health monitoring and treatment. These technologies enable the real-time tracking of vital health indicators, including heart rate, body temperature, respiration, activity levels, and hydration [48]. Wearable technology in animal care includes smart collar-mounted devices, reproductive monitor sensors, and ear tags. These biosensors play a vital role in advancing animal health management by continuously tracking activity, behavior, and physiological data, providing valuable insights into the animal's health and reproductive status. Worldwide, various animal-specific sensors are progressing through different stages of commercialization. While some health and diagnostic technologies are initially developed for human use, only slight modifications or limited testing have made them applicable in veterinary contexts [49]. Various studies have explored wearable technologies, nano biosensors, and advanced molecular diagnostic methods for detecting infectious diseases in cattle, along with efforts to catalog and compare these technologies based on their advantages and limitations in animal health management

[50,51]. A recent study explored the use of a wireless sensor/device to evaluate rumen pH dynamics and productivity in dairy cows fed diets that induce subacute ruminal acidosis (SARA). SARA has a significant impact on cow health, welfare, and productivity, resulting in economic losses for farmers. Advancements in these technologies have led to the development of a non-invasive, farmer-friendly wireless sensor capable of providing continuous monitoring of rumen health [52]. Biosensors integrated with cellphones and handheld devices simplify farm monitoring by replacing manual note-taking and basic tools that lack data-sharing ability. They enable real-time data collection, wireless transmission, and remote access, improving accuracy, efficiency, and timely decision-making in animal health management. Several systems have been developed specifically for cell phones and handheld devices to reduce the need for manual data recording [53]. Currently, biosensor devices available on the market are: Maven AI-Vet, an AI-based smart collar sensor that monitors detailed health parameters in pets; Kippy, a device providing GPS tracking for livestock; and ear tag sensors that monitor activity, temperature, and estrus health in animals [26,54]. Collectively, these wearable biosensors enhance animal health monitoring by providing real-time, actionable data that supports timely interventions and improves overall care. Additional wearable technologies have been developed to track more specific health indicators. In a recent study, a new wireless body electrode for long-term heart monitoring showed promise in veterinary applications, transmitting ECG data to a smart device via low-power Bluetooth. This device was tested on two dogs (one with sinus rhythm and another with atrial fibrillation and extrasystoles) and successfully identified different heart rhythms and conditions. Its user-friendly design, reliability, and extended battery life

suggest it could serve as an alternative to traditional Holter monitors in veterinary care. [55]. Wearable technology is also aiding in the management of chronic conditions in animals. For instance, diabetic animals benefit from continuous glucose monitors, enabling veterinarians to make precise adjustments to treatment plans based on real-time data. [56]. Wearables in veterinary care are transformative, offering real-time insights that promote efficient, preventative, and individualized care, ultimately enhancing animal health and quality of life.

3.3. Artificial intelligence in diagnostics

Artificial intelligence (AI) is revolutionizing veterinary diagnostics by enhancing the accuracy, speed, and efficiency of animal healthcare. AI-driven diagnostic systems, often integrated with sensors and wireless communication technologies, enable real-time and continuous monitoring of diverse health parameters in livestock. [57]. This technology can process vast amounts of data, including veterinary imaging, laboratory results, and animal medical histories, to assist veterinarians in making more accurate diagnoses [58]. Veterinary medicine, covering areas like companion animal health, population medicine, and agriculture, is rapidly evolving with AI. AI is increasingly being applied in veterinary practice for improving diagnostic accuracy and overall efficiency. With ongoing advancements in research, it holds significant promise to further enhance veterinary care, ultimately benefiting both animals and their caretakers in the future. [59]. In a previous study, a cattle rumen device utilizing advanced AI techniques for data processing operates across four stages, from measurement to integrated analysis [60]. The researchers concluded that meaningful outcomes in this area can only be achieved by applying modern computer science tools and intelligent data analysis at every stage. Also, a deep learning AI algorithm was developed to identify impacted bones in animals from lateral neck radiographs and evaluate its effectiveness in enhancing the interpretation of these images [61]. The AI model showed increased sensitivity for detecting bone impaction without raising the false positive rate, effectively reducing diagnostic time, speeding up workflows, and decreasing the reliance on CT scans. AI-driven smart devices for dairy cattle improve veterinary care by tracking key health indicators, including rumen function and feed consumption. By processing data in real-time, these devices help identify early signs of illness or stress, allowing for prompt action, enhancing milk yield, optimizing herd management, and ultimately improving farm productivity and animal well-being [62]. Managing livestock herds is a complex responsibility requiring specialized expertise, attention to animal health, and monitoring of worker performance. It involves analyzing diverse data to make informed decisions. As a result, herd management systems, automation, and AI have become essential tools for enhancing farm sustainability and profitability. The study examines the evolution of these technologies in modern animal husbandry, tracing their development from early adoption to current applications and highlighting the benefits they provide. [63]. A previous study reported a cattle rumen device that applies advanced AI techniques for data processing through four stages, from measurement to integrated analysis. The researchers emphasized that significant progress in this field requires the use of modern computer science tools and intelligent data analysis at each stage. technology provides real-time insights, which are critical for prompt interventions. The system alerts veterinarians or farmers when specific thresholds are met, enabling timely action to prevent health issues from worsening. Additionally, this technology helps prevent the overuse or misuse of antibiotics by allowing more accurate monitoring of animal health and behavior. As AI continues to advance, it holds great potential for improving diagnostic accuracy, disease prevention, and overall animal health and productivity in veterinary care.

3.4. Remote consultations in veterinary care

Remote consultations utilizing smart technology are transforming

veterinary care by enabling veterinarians to remotely monitor and manage livestock health through real-time data collected by ingestible devices [64]. Telehealth, a swiftly advancing field in veterinary medicine, encompasses various technologies to deliver health information, education, and care remotely. Teleconsulting, a subset of telehealth, involves collaboration between an on-site veterinarian and a remote consultant, maintaining high standards of care while enabling physical distancing [65]. Animal-human relationships, which offer mutual benefits, are strengthened through the use of proper treatments that relieve stress and discomfort. The long-acting formulations play a crucial role in treating animal diseases, improving compliance, and reducing caregiver burdens, paving the way for better, sustainable care for chronic conditions and advancing the future of veterinary healthcare [66]. Telemedicine-enabled devices are revolutionizing veterinary care by allowing remote health monitoring of livestock and delivering real-time data to veterinarians and farmers [67]. These devices enhance farm management by enabling continuous monitoring of livestock health and activity, allowing farmers to fine-tune feeding schedules, track reproductive cycles, and maintain overall animal well-being. In dairy farming, these devices help monitor milk production levels, optimizing yield while keeping animals healthy and stress-free [68]. By reducing the need for frequent physical check-ups, telemedicine devices cut labor and time costs associated with veterinary care, providing accurate and convenient health insights [69]. This technology is invaluable for modern veterinary practices, supporting more sustainable and profitable farming operations.

4. Smart bolus and its mechanisms in veterinary healthcare

A smart veterinary bolus is an advanced oral device administered to livestock typically ruminants such as cattle, sheep, or goats, that goes beyond the function of a traditional bolus. Unlike conventional boluses, which primarily deliver drugs, minerals, or nutrients, the smart boluses integrate controlled-release systems and sensors for precision delivery and real-time health monitoring. Functionally, smart boluses are classified into two main types: drug/nutrient delivery boluses that employ controlled-release mechanisms for long-term administration of medications, minerals, or hormones. Sensor Boluses include temperature-sensing, pH-sensing, activity, and motion sensor boluses that monitor internal parameters such as rumen physiology, pressure, activity, and biomarkers. These are applied for disease detection, digestive health assessment, and other physiological monitoring. The mechanistic role of smart boluses in veterinary care involves the integration of controlled-release systems, advanced biosensors, data analytics, and wireless communication. Data collected from these biosensors is transmitted wirelessly to a central system, enabling real-time monitoring by veterinarians and farm managers. [70]. This information offers valuable insights into the animal's health, enabling prompt action and targeted care. The mechanism of the smart bolus is described below.

4.1. Smart bolus for controlled release systems

The mechanisms of smart bolus systems in controlled release systems for veterinary care represent a significant advancement in the targeted and sustained delivery of therapeutic agents within livestock. Table 2 illustrates the comparative analysis of different controlled drug delivery systems used in animal bolus applications. These systems are engineered to release active compounds, such as drugs, vitamins, anthelmintics, antibiotics, hormones, growth alterants, minerals, trace elements, and nutrients, at a controlled rate, ensuring consistent therapeutic levels over extended periods [71,72]. This formulation and technology, featuring a controlled release system, is often triggered by environmental factors within the animal's gastrointestinal tract, such as enzymatic activity, allowing for site-specific delivery. The recent study investigated controlled-release intraruminal boluses (CRIRB) as a means of delivering essential trace elements to ruminants, which are crucial for

Table 2
Comparison of different controlled drug delivery systems used in animal bolus technology.

Controlled Release System	Mechanism	Advantages	Limitations	Suitability for Livestock
Diffusion-Controlled System	Drug diffuses through a polymer matrix or membrane into surrounding fluids, driven by a concentration gradient.	Simple design, sustained release, relatively low cost.	Release rate is affected by pH, temperature, and fluid flow in the rumen; less precise over long durations.	Suitable for delivering minerals, vitamins, or drugs requiring steady release over days–weeks.
Osmotic-Controlled System	Semi-permeable membrane allows water influx, creating osmotic pressure that pushes dissolved drug through a small orifice at a controlled rate.	Highly precise and consistent release, independent of rumen environment.	More complex and costly to manufacture; potential membrane clogging.	Ideal for long-term delivery of drugs or hormones in cattle, sheep, and goats.
Erosion-Controlled System	A drug embedded in a biodegradable matrix is released as the material erodes in digestive fluids.	Eliminates retrieval needs; adjustable release rate via matrix composition.	Release rate may vary with digestive conditions; material selection is critical.	Useful for single-course treatments or nutrient supplementation without bolus recovery.
Microchip-Controlled System	Embedded microchip triggers drug release at programmed intervals or in response to physiological/environmental signals.	Highly programmable, can adapt dosing to real-time conditions; integrates with livestock monitoring systems.	High cost, requires robust electronics to survive rumen conditions, and battery limitations.	Best for high-value livestock or research, precision dosing, and targeted disease management.

sustaining life processes and promoting growth. It further examined different types of CRIRB designed for trace mineral supplementation and evaluated various manufacturing approaches, including hot-melt

extrusion, melt granulation, and direct melting techniques. [73]. Additionally, a controlled-release albendazole capsule (CRC) was administered to sheep as a model for worm-free animals in grazing trials. The

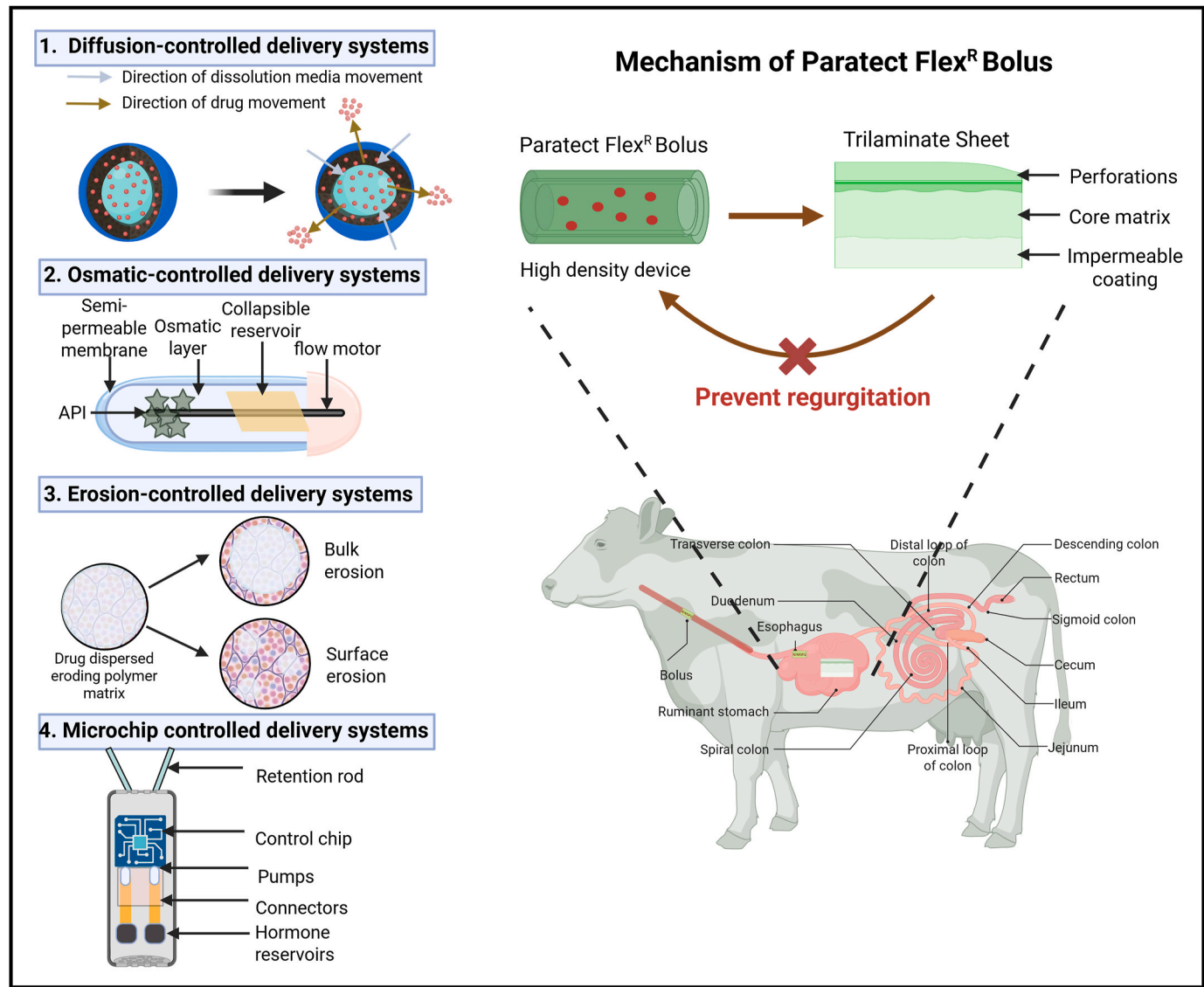


Fig. 2. Mechanisms of drug release from veterinary drug delivery systems: (A) Different types of boluses for controlled delivery systems; (B) Paratect® Flex Bolus, a high-density device that transforms from a cylindrical form into a trilaminar sheet, preventing regurgitation in cows (Created through [BioRender.com](#)).

results of this study indicate that, although CRC exhibits mild immunogenic properties, it does not lead to production loss. The treatment reduced live weight gain, and administering anthelmintic treatment at a worm egg count (WEC) threshold of 1500 eggs per gram (epg) was found to minimize production losses during infections and prevent lingering adverse effects after infection [74]. Apart from this, naltrexone (NLX), an opioid antagonist, faces challenges in oral administration due to poor bioavailability and patient adherence [75]. In Fig. 2, the Paratect Flex bolus, composed of ethylene vinyl acetate with morantel tartrate, is rolled into a cylinder coated with a water-soluble film. After throat administration, it unrolls to prevent regurgitation [76,77]. However, the development of such systems presents challenges, including ensuring biocompatibility, optimizing the release kinetics for different species and conditions, and maintaining the structural integrity of the bolus in the harsh gastrointestinal environment. The integration of these smart mechanisms into veterinary care has the potential to significantly improve the efficiency and effectiveness of animal health management.

4.2. Smart bolus for biomarker monitoring in veterinary healthcare

Biomarker-based boluses/devices represent a major advancement in veterinary care, providing precise diagnostic and therapeutic functions through continuous, real-time monitoring of animal health within the gastrointestinal tract. These smart systems detect key biomarkers, such as enzymes and metabolic indicators, associated with inflammatory, infectious, or metabolic disorders, and can administer medications, nutrients, or trace elements in response to specific health conditions. By delivering targeted treatment only when needed, they enhance therapeutic effectiveness, improve outcomes, and reduce the necessity for repeated dosing [78]. The collected data is wirelessly transmitted to external platforms, enabling veterinarians and farmers to monitor health trends, identify diseases at an early stage, and implement timely management strategies. Through such real-time, personalized interventions, this technology supports improved livestock health, higher productivity, and enhanced welfare [79]. In parallel, intensification of milk production has increased the incidence of metabolic diseases in high-yielding dairy cows, negatively affecting their health and productivity. Biomarkers found in plasma and milk, including indicators of energy balance, oxidative stress, fertility-related hormones, and disease-specific metabolites, offer promising tools for the early detection of these disorders, supporting more effective management and animal welfare [80]. Ultimately, these biomarker-integrated boluses are transforming veterinary medicine by improving diagnostic accuracy, streamlining treatments, and promoting more precise and welfare-driven animal care.

4.3. Smart bolus with monitoring equipment and feedback loops

Smart bolus sensors with monitoring and feedback loops are revolutionizing veterinary care by improving real-time health management, particularly in large-scale livestock operations. These technologies enable the continuous monitoring of vital signs, behavioral changes, and environmental factors, allowing veterinarians and farm managers to identify health issues early and take proactive measures [27,81]. Technological innovations are reshaping livestock management, with smart boluses equipped with sensors to monitor key physiological and health parameters such as body temperature, pH levels, and rumen activity. This study underscores the importance of continuous advancements in animal health technology [82]. Monitoring the health of dairy cattle is critical for increasing milk production. A recent study explores wireless sensor network (WSN)-based systems for health tracking and disease detection, enabling farmers to remotely monitor cattle, store data for analysis, and optimize farm management. The low-cost WSN device enhances productivity while minimizing human intervention, addressing challenges in dairy farm automation [83]. These systems utilize sensors to measure various physiological parameters, transmitting data wirelessly to a central system where advanced algorithms

analyze the information and provide actionable feedback. This feedback loop allows for real-time adjustments in treatment protocols, such as medication dosages or feeding strategies [84]. The smart rumen bolus sensor by Moonsyst, shown in Fig. 3, exhibited a potential lifespan exceeding six years. This durability enables long-term monitoring of body physiology and activity in livestock, as confirmed through battery performance evaluations. The bolus represents a reliable solution for efficient, extended animal health and behavior tracking in veterinary practices [79,85]. The incorporation of monitoring and feedback systems improves the accuracy of veterinary interventions while increasing herd management efficiency by minimizing manual inspections and allowing for prompt action on emerging health concerns.

5. Different methods of administration of smart boluses and devices

In veterinary healthcare, the path or delivery route of bolus/device administration is primarily employed for the delivery of smart sensors, API, and nutrients to livestock, with the choice of administration route determined by the species of the animal, the specific condition, and the intended therapeutic outcome [86]. Moreover, several active pharmaceutical ingredient (API) dosages and delivery methods have been developed to improve animal health and welfare. In veterinary medicine, dosage forms are typically formulated by combining an API with one or more excipients. The resulting formulation determines the administration route and significantly influences the therapeutic efficacy and safety [87]. Fig. 4 depicts the various delivery methods for smart boluses.

5.1. Oral administration

Oral administration of boluses is one of the most common and convenient methods for drug delivery, particularly for routine and long-term treatments in livestock. This route allows medications to travel through the digestive tract and be absorbed into the circulatory system. Boluses are commonly used in the form such as magnets, antibacterial sulfa drugs, coccidiostats, slow-release nutritional boluses (e.g., copper boluses, vitamin mixes, rumen-stimulating yeast), and antiparasitic medications [73]. Their administration requires a bolus or balling gun; a long cylindrical tube typically made of stainless steel or plastic. The gun may feature either a spring-action handle or a three-holed plunger, which functions similarly to a syringe, but with a thumb-operated top and two-finger grips. This design pushes the bolus deep into the cow's mouth to promote swallowing [6,88]. In a previous study, researchers evaluated the impact of calcium bolus supplementation on early-lactation health and milk yield in dairy cattle. They divided cows into two groups: a control group that received no calcium boluses around calving, and a treatment group that received two oral calcium boluses. The treatment cows were given the first bolus within 0–2 h post-calving and the second between 8 and 35 h after calving. The results showed that cows receiving calcium boluses produced an additional 2.9 kg of milk at their first post-calving test compared to cows with higher previous lactation milk yields that did not receive supplementation. The study concluded that oral calcium bolus supplementation is beneficial for specific subgroups of cows, even in herds with low rates of hypocalcemia [89]. This method is typically non-invasive and appropriate for a variety of conditions, including chronic illnesses and nutritional supplementation. However, factors like gastrointestinal conditions, inconsistent absorption rates, and possible breakdown of the drug by digestive enzymes can impact its effectiveness. Additionally, the onset of action is often slower compared to other methods, making it less suitable for acute situations where rapid intervention is needed.

5.2. Intravaginal administration

The estrous cycle in farmed animals is managed using intravaginal

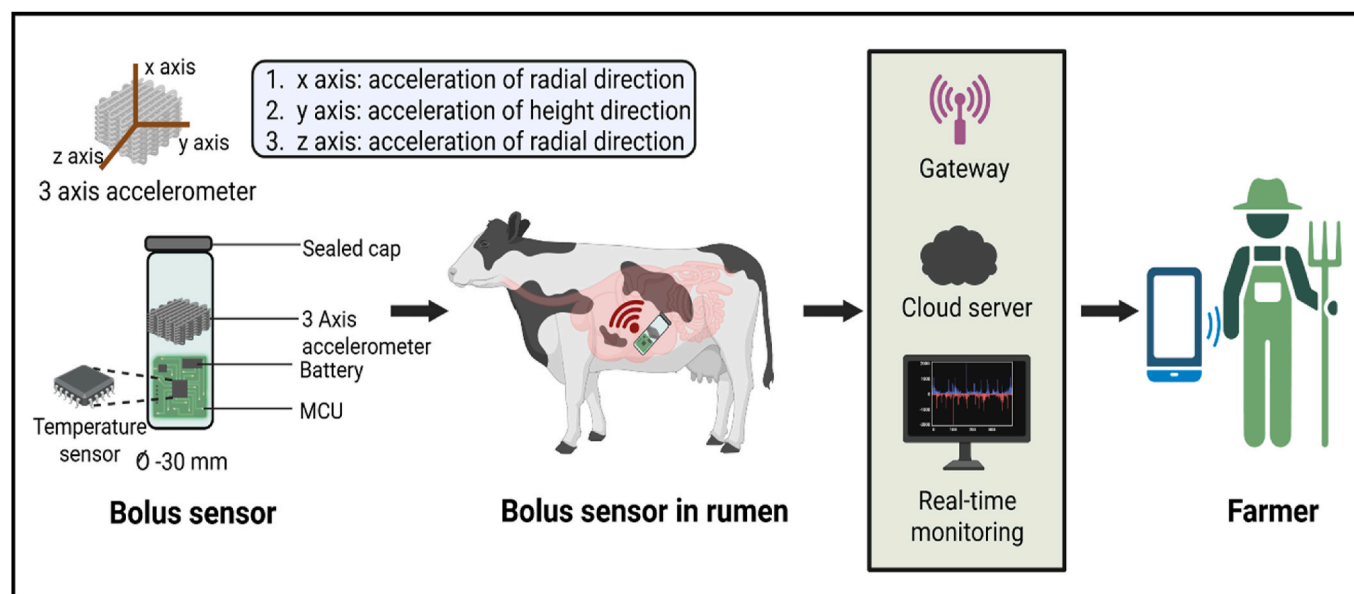


Fig. 3. A model of a smart rumen bolus node by Moonsyst (courtesy of Moonsyst) and a schematic diagram of an Internet of Things (IoT) sensor system for real-time monitoring of the rumen environment (Created through [BioRender.com](https://www.biorender.com)).

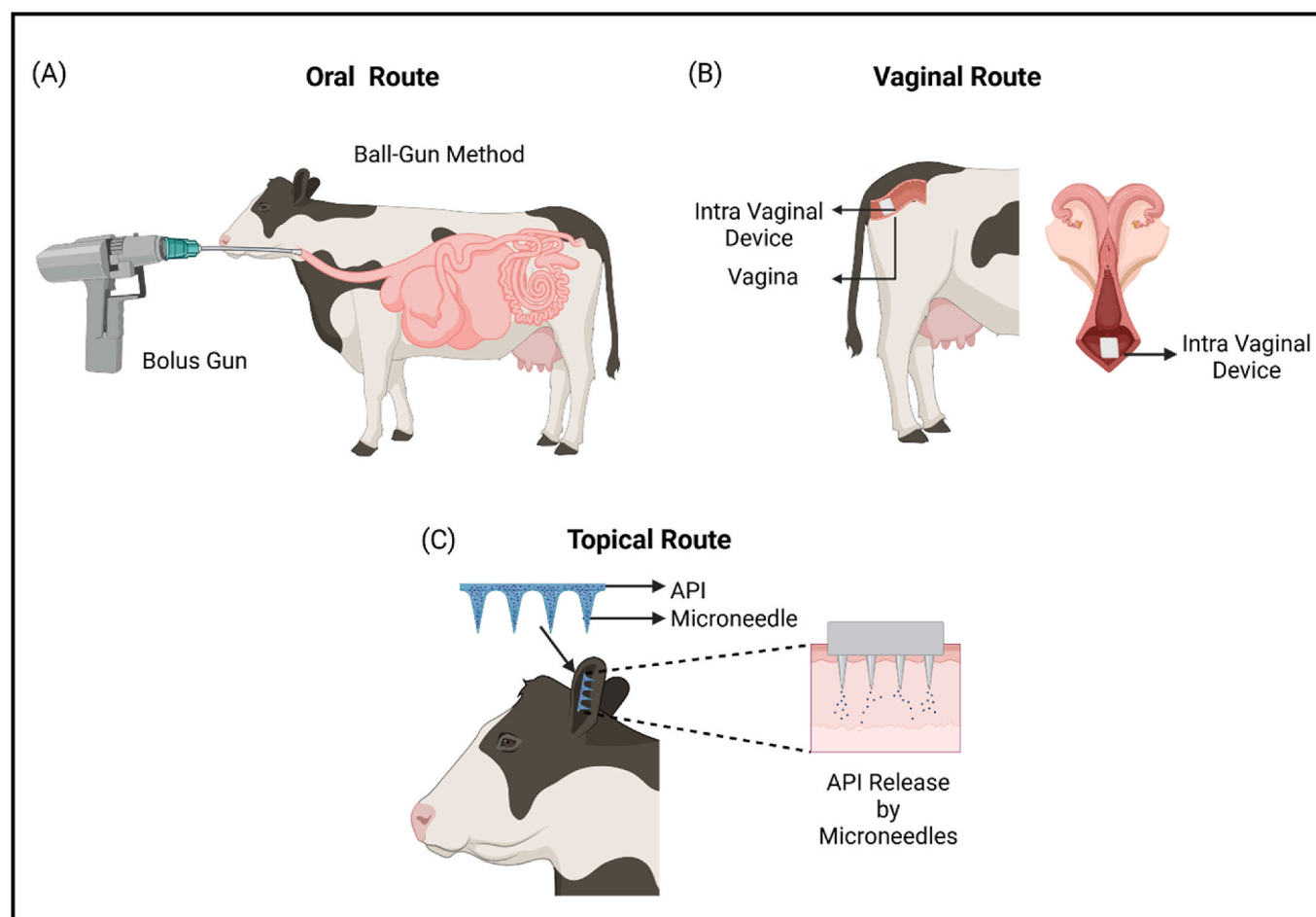


Fig. 4. Various routes of administration for veterinary medicine: (A) Oral delivery of a bolus using the ball-gun method, (B) Intra-vaginal devices for administering medicines related to animal reproductive health, and (C) Microneedles for topical drug delivery (Created through [BioRender.com](https://www.biorender.com)).

devices designed with a silicone-progesterone matrix. Commercially available intravaginal devices include Controlled Internal Drug Release (CIDR) variants such as CIDR-B, CIDR-G, and CIDR-S, as well as Progesterone-Releasing Intravaginal Devices (PRID). This method is particularly favored for sheep and goats and was later adapted for cattle. The silicone particles within the device ensure a controlled and gradual release of progesterone, prolonging its effect *in vivo*. Additionally, silicone's remarkable heat stability makes it ideal for embedding progesterone in these devices [90]. In a past study, Rathbone and colleagues investigated the application of poly(ϵ -caprolactone) (PCL) as a substitute for silicone in intravaginal devices. Unlike silicone, which is non-biodegradable, PCL is a partially crystalline synthetic polyester that degrades at a rate of 20 % annually. PCL is compatible with intravaginal use, causing no burning sensation. Comparative studies revealed similar retention rates for PCL (99.7 %) and CIDR (97.2 %), although CIDR exhibited higher plasma progesterone levels. Speculum-assisted vaginal examinations confirmed that both devices deliver bioequivalent outcomes, highlighting PCL's suitability for controlled progesterone release [91]. The researchers also investigated the use of intravaginal devices/boluses for probiotic delivery in sheep. Probiotics, such as *Lactobacillus* species, offers a natural alternative to antibiotics. Bolus/Sponges enriched with Lactic Acid Bacteria (LAB) were found to mitigate inflammation and hemorrhage associated with vaginal discharge while effectively preventing infections, supporting their potential benefits in sheep reproductive care [92].

5.3. Other routes of administration

The choice of administration route is influenced by various factors, including the drug's pharmacokinetics, the target condition, the desired speed of action, and the animal's specific needs. Topical administration involves applying a synthesized formulation directly to the skin or mucous membranes, offering localized treatment for dermatological conditions, parasitic infections, or certain types of pain relief [93]. Topical therapy is an important part of veterinary dermatology; it is now recognized as a key component of the management of bacterial and parasite overgrowth (often as the first line of therapy), primary seborrhea, and allergic dermatitis [94]. A recent study discussed various topical devices, including cattle ear tags, different types of collars, and bioavailable microneedles designed for controlled release at infection sites in animals. Researchers are particularly interested in exploring less-studied delivery systems, such as topical delivery systems, for this purpose [95]. Apart from this, microneedles represent an innovative technology for delivering APIs via the topical route. Microneedle patches offer a painless method for transdermal drug delivery. *In-vivo* studies confirmed swelling, dissolution, and no adverse reactions, highlighting their potential for effective cattle pain management and commercial production [96]. Additionally, researchers developed and assessed chitosan microneedle patches for the transdermal delivery of meloxicam to alleviate pain in cattle. The patches, prepared with 10 %, 50 %, and 90 % acetic acid concentrations, were evaluated for chemical properties, uniformity, skin penetration, and thermal resistance. They achieved a penetration depth of $78 \pm 1 \mu\text{m}$ and released 26.2 % of the required drug concentration weekly, demonstrating their potential for cattle pain management [97]. Each route of administration for veterinary products has unique benefits and limitations that require careful consideration to ensure effective therapeutic results. Ongoing advancements in veterinary medicine continuously improve the development and use of these administration methods, contributing to better health and welfare for animals under veterinary care.

6. Applications of advanced technology in veterinary medicine

Smart technology is a major step in precision livestock management and animal health monitoring, and care. These smart devices are designed to overcome limitations of conventional treatment systems,

continuously monitor key physiological parameters within the animal. [98]. Specific devices/boluses are developed to monitor physiology and pathology in ruminants such as calves, sheep, goats, and deer. These devices are particularly suited for these animals, offering distinct advantages over larger products currently available on the market. Likewise, the smart rumen bolus captures data on animal activity, rumen temperature, and movement patterns, transmitting this information to a gateway over an extensive geographic area [99]. One significant application of this technology is in managing chronic conditions, such as diabetes in companion animals or metabolic disorders in livestock, as smart devices allow for precise medication delivery and continuous monitoring of treatment efficacy. In addition, they support optimized nutritional management by evaluating digestive efficiency and nutrient absorption in ruminants, enabling more individualized feeding approaches and improved growth outcomes [100]. Smart technology is also invaluable in early disease detection; shifts in monitored parameters can signal health issues before visible symptoms arise, facilitating timely interventions and reducing the likelihood of severe outbreaks [101]. Despite the advantages, challenges such as ensuring device resilience, data reliability, and compatibility with current veterinary practices remain. However, with ongoing development, smart technology has strong potential to transform veterinary medicine by improving diagnostic precision and treatment success rates.

6.1. Health monitoring

Novel technology within veterinary medicine offers enhanced real-time animal health surveillance capabilities. These devices, ingested by the animal, utilize embedded sensors to continuously track vital physiological parameters such as ruminal physiology and heart rate [102]. A different study demonstrated that emerging technologies were utilized to understand animal health and disease. The latest study utilized advanced therapeutic diagnostic tools, including next-generation boluses/devices, to address the limitations of traditional antibiotic therapies [103]. Additionally, smart sensor technology has been used to gather data on animal activity and rumination. Marian and colleagues applied this technology for detecting estrus and clinical mastitis (CM) on two dairy farms in northeastern Germany, demonstrating its effectiveness in monitoring reproductive and health conditions in dairy cattle [104]. Invasive wearable sensors, including lumen and vaginal devices, are available for tracking physiological and behavioral changes essential for diagnosing animal health issues. Scientists have developed two types of smart sensors for reproductive health: one monitors body surface temperature at the ventral tail base, and the other detects estrus in cattle to improve health and reproductive monitoring [105]. Some calving detection sensors are designed for direct insertion into a cow's vagina. These smart devices monitor body physiology, triggering a calving alert based on the typical temperature drop before birth. Another type of vaginal smart device detects light exposure; once expelled by the cow during a water break, the sensor recognizes it's outside the body and sends a notification to farm managers, signaling calving has begun [100, 106]. This technology is especially useful for early disease detection, as it identifies subtle physiological changes that can appear before visible clinical symptoms, allowing for prompt intervention. Additionally, smart boluses/devices provide essential data for managing chronic conditions in companion animals, like diabetes or heart disease, by continuously monitoring and assessing treatment responses. Ongoing advancements in smart technology are expected to expand its role in comprehensive health monitoring, ultimately promoting better animal welfare and management efficiency.

6.2. Drug delivery

In drug delivery, smart device/bolus technology represents a revolutionary advancement in veterinary medicine, enabling precise and controlled therapeutic interventions. These ingestible formulations and

devices can be tailored to deliver medications to specific sites within the gastrointestinal tract, utilizing a controlled release mechanism [107]. Also, these boluses incorporated nanoparticles, targeting antibiotic-resistant strains and supporting ethno-veterinary approaches for treating mastitis in dairy cattle [108]. A previous study highlighted that novel drug delivery systems (NDDSs) can be developed by veterinarians to address various medical conditions in horses effectively. This approach considers essential factors for delivering oral, topical, and parenteral drugs, accounting for age and species differences, as well as potential challenges related to dosage forms specific to horses [109]. Notably, the controlled release mechanism ensures that the animal receives a consistent supply of these essential nutrients over time, which promotes improved health and productivity. One notable application of this technology is in delivering long-acting medications, like hormones or antibiotics, where a smart bolus can be programmed for extended release, maintaining steady therapeutic levels and reducing administration frequency. In another study, researchers investigated controlled metronomic drug delivery to the brain using an implantable smart pump in a sheep model. This implantable pump system allows external regulation of drug release, maintaining intraventricular drug levels consistently within the therapeutic range [110]. Apart from this, the preparation and *in vitro* release kinetics of an ivermectin (IVM) sustained-release bolus, optimized using response surface methodology, have been studied for treating parasitic diseases in animals. This bolus, composed of microcrystalline cellulose (MCC), starch, and low-substituted hydroxypropyl cellulose (LS-HPC), was dissolved in 900 mL of dissolution medium at 39.5 °C with stirring at 100 rpm, demonstrating a diffusive erosion release mechanism [111]. The continuous advancement of smart bolus technology holds great potential for improving the precision and effectiveness of drug delivery in veterinary medicine, offering customized therapeutic solutions for various animal health challenges.

6.3. Nutrient supplementation

Another application of advanced technology in nutrient supplementation for veterinary medicine, facilitating the precise and efficient delivery of essential nutrients to livestock and companion animals. Boluses with supplements frequently include vitamins, minerals, and other nutrients that are vital for the well-being of the animal. The nutritional supplement contains vital microminerals, such as copper and zinc, to support estrus, boost breeding efficiency, preserve a healthy pregnancy, and guarantee normal delivery in animals [112]. A previous study explored how nutrition and precision technology can enhance health and performance and reduce stress in beef cattle. Efficient beef production systems are essential to boost productivity with fewer resources, promoting sustainability and minimizing environmental impact. In this context, a smart farming approach validated the use of an automated scale system for grazing or feedlot beef cattle, which accurately measured body weight in both growing and mature cattle, reducing the need for frequent handling [113]. Smart bolus/device enables precise, sustained delivery of nutrient supplements, improving livestock health and productivity. A recent study investigated innovative nutrient delivery systems for beef cattle by evaluating the efficacy of a sustained-release mineral-based bolus. The research comprised two separate experiments conducted in a completely randomized design using 16 ruminal cannulated cows. The first experiment assessed the degradation rates of two bolus prototypes over 87 days in 2-year-old and 3-year-old cows. The second experiment compared the performance of boluses with targeted degradation durations of 90 and 180 days under two dietary conditions: low-quality, high-fiber forage (>600 g/kg neutral detergent fiber [NDF], <80 g/kg crude protein [CP]) and high-quality, low-fiber forage (<500 g/kg NDF, >150 g/kg CP). This work aimed to elucidate the degradation behavior of boluses and their interactions with forage quality. [114]. Notably, the controlled release mechanism ensures that the animal receives a consistent supply of these

essential nutrients over time, which promotes improved health and productivity. Ingestible boluses are designed to deliver controlled doses of vitamins, minerals, and other supplements directly to the gastrointestinal tract over extended periods. This sustained release mechanism provides animals with a steady supply of nutrients, helping to manage deficiencies and promote optimal growth and health in livestock. Reticulorumen boluses are also utilized to assess milk production levels and trace mineral profiles in cows. The impact of bolus administration on body weight change during early lactation varies depending on breed and year, suggesting a potential benefit in developing breed and year-specific supplementation strategies [115]. This treatment resulted in higher early milk yield and a trend toward greater income compared to the control, though with lower colostrum protein and amino acid levels [116]. Furthermore, they are used as carriers for vitamins, essential minerals, and trace elements such as iodine, cobalt, and selenium, which are crucial for the growth, reproduction, and overall health of livestock [117]. Additionally, hormonal boluses/devices are employed to manage reproductive health by regulating estrous cycles, treating hormonal imbalances, and addressing conditions such as retained placenta and metritis in dairy cattle [118]. These interventions are crucial for maintaining reproductive efficiency, which is essential for the economic sustainability of livestock farming. Moreover, this technology allows for customized nutrient delivery, which can be adjusted in real time based on the animal's physiological status and specific health or production needs. Smart technology also reduces the labor demands and frequency of traditional supplementation, enhancing efficiency in large-scale farming operations. Ongoing advancements in smart bolus technology are expected to further improve the precision and effectiveness of nutrient supplementation, supporting better overall animal health and productivity.

6.4. Behavioral monitoring

Smart technology plays a crucial role in veterinary behavioral monitoring, enhancing livestock health management through real-time tracking and data-driven insights. These ingestible devices monitor core heart rate, rumen pH, motility, and activity levels, providing veterinarians with early indicators of health issues like fever, digestive disorders, or stress-related behavior [98,119]. These ingestible devices are equipped with advanced sensors that can continuously monitor various physiological and environmental parameters, such as movement patterns, rumination, and feeding. As climate change drives the need for livestock heat stress detection, non-invasive methods are evolving, focusing on behavioral and physiological indicators without additional stress on animals. Technologies like bioclimatic indices, fecal microbiome analysis, and cortisol measurements from various devices, along with tools such as infrared thermography, accelerometers, GPS, and machine learning, offer valuable insights into livestock welfare and resilience for sustainable production [120]. The wireless analyzing data from these devices, veterinarians and caretakers can detect behavioral deviations, potentially signaling stress, illness, or environmental changes affecting welfare [121]. While advanced technology offers significant benefits, ongoing developments are needed to enhance data accuracy and minimize impacts on natural behavior, ultimately improving welfare and management practices.

7. Future Directions, industrial opportunities, and challenges

Emerging technologies in veterinary care are set to transform livestock health management, presenting considerable industrial opportunities. As technology advances, several promising areas for innovation and development are coming into focus. Technologies like Smart sensor, artificial intelligence (AI), the Internet of Things (IoT), 5G, and robotics are creating new pathways for rapid communication, extensive connectivity, and data-driven decision-making, paving the way for truly connected farms of the future [122,123]. Similarly, veterinary medicine

is increasingly adopting advanced technology to reduce human error. Most studies focusing on detecting animal activity, behavior, and physiological changes employ off-body technologies, providing continuous data on behavior and health metrics over time. Strong collaboration between veterinarians and animal-technology interaction specialists will further enhance and refine medical devices designed specifically for animals [124,125]. The key advancement lies in the integration of artificial intelligence (AI) and machine learning with advanced technology, enabling predictive analytics for a range of applications in animal healthcare (Fig. 5). By analyzing data collected from AI, smart devices can identify patterns and predict health issues, allowing farmers and veterinarians to detect illnesses early and apply timely interventions [126,127]. Additionally, AI can optimize treatment protocols dynamically, adjusting dosage or frequency based on the animal's real-time response, which could significantly improve the efficacy of therapeutic interventions. This shift from reactive to proactive health management can reduce disease incidence and improve animal welfare, ultimately leading to greater productivity and cost efficiency for livestock operations [128].

Furthermore, innovations in advanced device/bolus materials and design, such as using biodegradable or more robust components, could enhance the longevity of boluses and lessen their environmental footprint. These improved designs may also enable the creation of multi-functional devices/boluses that combine drug delivery, nutrient provision, and real-time health monitoring within a single device [123]. Another significant trend in smart bolus technology is the development of more sensitive sensors [129]. These sensors will monitor various biomarkers that reflect metabolic health, immune function, and stress levels, offering a detailed overview of an animal's well-being [98,130]. With more accurate data on these factors, veterinary professionals and livestock managers can make better-informed decisions regarding nutrition and disease management. In the industrial realm, opportunities for smart/advanced technology are rapidly expanding. Advancements in technology and reductions in production costs are making these systems more accessible to both large and small farming operations. The potential in sectors such as dairy, beef, and large-scale livestock farming is substantial, where animal health is crucial for productivity and profitability [131]. Veterinary pharmaceutical companies, agrotech firms,

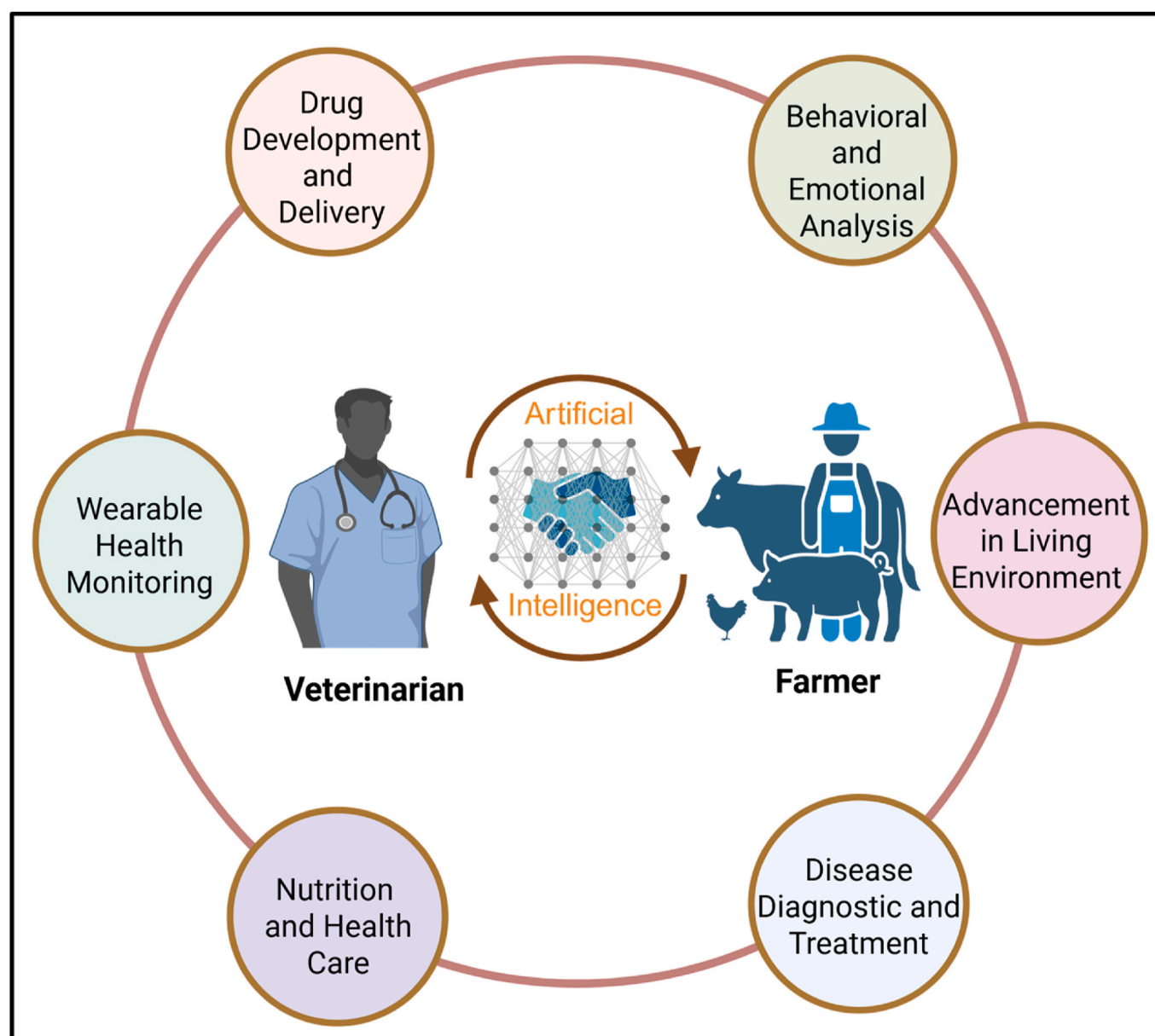


Fig. 5. A representation of various applications of Artificial Intelligence (AI) in animal healthcare (Created through BioRender.com).

and livestock management businesses have valuable opportunities to develop and market smart bolus products that meet the growing demand for data-driven and sustainable agricultural practices. Also, collaboration between tech companies and agricultural stakeholders can drive innovation in bolus/device design, broadening their market presence and applications.

Moreover, advancements in regulatory frameworks and quality standards for smart technology are expected to create opportunities for companies to collaborate with authorities in establishing industry benchmarks. As adoption of smart boluses grows, their benefits, ranging from improved medical outcomes and enhanced herd health to more efficient farming practices, are becoming increasingly evident. These innovations have the potential to redefine veterinary care, making livestock management more efficient, sustainable, and profitable, while meeting the rising demand for eco-friendly food production. With continuous technological progress, smart bolus systems could transform veterinary medicine by improving diagnostic accuracy, enabling personalized treatments, and streamlining health monitoring. Despite these potential, smart bolus technology in veterinary care faces notable challenges. High upfront costs and ongoing maintenance can limit adoption, particularly among small-scale farms. Technical constraints such as sensor calibration, battery life, and reliable data transmission also require attention. Data privacy, ownership concerns, and integration with existing farm management systems further complicate implementation. Additionally, the lack of standardized regulations and quality benchmarks hinders widespread use. Training veterinarians and farmers to effectively operate and interpret these systems is essential to unlocking their full potential. Ultimately, advanced technology and smart boluses/devices offer a pathway to enhanced disease prevention, more effective treatments, and improved animal welfare, positioning them as vital tools in the future of veterinary care.

8. Conclusion

Advancements in new technology are revolutionizing veterinary medicine by combining precise drug delivery with continuous health monitoring. These innovative boluses/devices integrate sensors, wireless connectivity, and biodegradable materials, enabling controlled administration of therapeutic agents while providing real-time physiological data, such as temperature and pH levels, directly from within the animal. This dual-purpose functionality is especially valuable in livestock and equine management, where reducing handling and ensuring consistent treatment are essential. Despite its benefits, including accurate dosing and improved therapeutic outcomes, the widespread adoption of smart technology is hindered by challenges such as high production costs, stringent regulatory requirements, and the need for extensive safety testing. Overcoming these obstacles will require ongoing research to enhance device durability, biocompatibility, and seamless data integration. With advancements in materials science, sensor technology, and data analytics, smart boluses, such as 3D-printed devices, can revolutionize veterinary care, improve animal health, and promote sustainable farming practices globally.

CRedit authorship contribution statement

Agri Jhila: Writing – original draft, Investigation, Data curation. **Krishna Jadhav:** Writing – original draft, Investigation, Data curation. **Raghuuraj Singh:** Methodology, Data curation. **Swarnima Negi:** Methodology, Data curation. **Shaguneet kaur:** Methodology, Data curation. **Neelesh Sharma:** Writing – review & editing, Validation, Resources. **Rahul Kumar Verma:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

3D: 3 dimensional, AI: Artificial intelligence, BCS: Body condition score, WDA: Weight per day of age, API: Active pharmaceutical ingredients, CRD: Completely randomized design, WSN: Wireless sensor network, FFF: Fused filament fabrication, CRIRB: Controlled-release intraruminal boluses, CRC: Controlled-release albendazole capsule, WEC: Worm egg count, NLX: Naltrexone, EG: Endothelial glycocalyx, SDG: Sustainable development goals, UK: Urokinase, CM: Clinical mastitis, NDDS: Novel drug delivery systems, IVM: Ivermectin, MCC: Microcrystalline cellulose, SARA: Subacute ruminal acidosis, GPS: Global positioning system, IoT: Internet of Things.

Data availability

Data will be made available on request.

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